Qatar-2b

R-1859

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-2_150328` · created 2026-08-03T09:08:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457109.803583 +/- 7.4e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.012
Transit depth (Rp/Rs)^2	2.2 +/- 0.37 [%]
Orbital Inclination (inc)	89.73 +/- 1.22
Airmass coefficient 1 (a1)	0.98 +/- 0.015
Airmass coefficient 2 (a2)	0.0161 +/- 0.0078
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#3 - [288, 133]
Optimal Aperture	2.99
Optimal Annulus	9.24
Transit Duration (day)	0.0757 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.012 vs 0.16327 (deviation 1.27σ)
Duration fit vs published	0.0757 d vs 0.0767 d (deviation 0.81σ)
Mid-time O–C	-0.0 min
Depth SNR	12.3
Residual scatter	1.51 %

Light curve

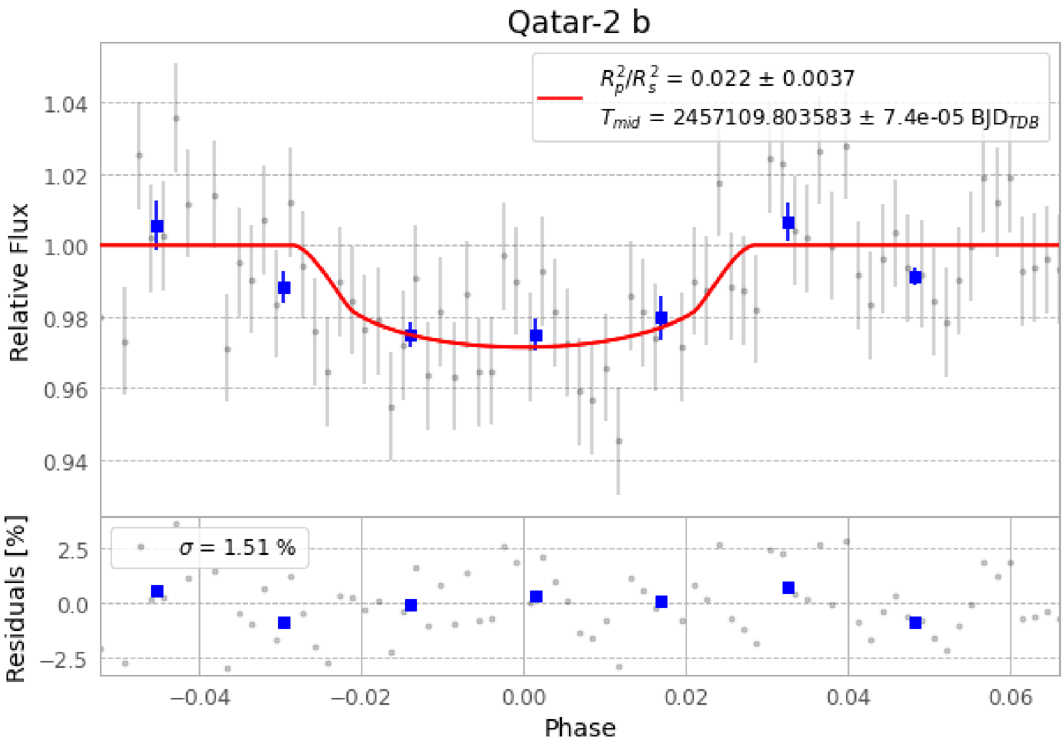

Figure 1 — FinalLightCurve_Qatar-2 b_2015-03-27.png (SHA-256 ec8c6043edb0e251…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [74, 208], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 26d8145cda98d656…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `Qatar-2150328052712.FITS` → `Qatar-2150328091512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-150328.log (490e7f27652e…), FinalLightCurve_Qatar-2 b_2015-03-27.png (ec8c6043edb0…), FinalLightCurve_Qatar-2 b_2015-03-27.csv (83edd688bebc…)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)